Sales Forecasting & Demand Intelligence — Business Report

Prepared for: Head of Supply Chain & CFO

Executive Summary

I built a system that looks at four years of our sales data and tries to answer one simple question: how much are we going to sell next, and what should we watch out for? I tested three different ways of forecasting sales and picked the one that gave the most accurate results. I also built a tool that flags unusual sales weeks, and a way of grouping our products by how they behave in the market, so we know which ones need close attention and which ones we can leave alone. The short version: our sales are growing overall, they follow a clear seasonal pattern every year, and I can now forecast the next three months with a fair degree of confidence, flag the weeks that don't follow the usual pattern, and tell you which product groups need tighter stock control.

Key Findings from the Data

- Technology is our highest revenue category overall, and the East region has shown the most steady, year-over-year sales growth of all four regions.
- On average, it takes about 3 days between when an order is placed and when it ships. This is consistent across regions, except the Central region, which takes about 4 days.
- Our sales follow a clear yearly pattern: March, September, and November consistently see higher sales every single year. This lines up with typical spring buying, back-to-school demand, and pre-holiday shopping.
- Overall, sales are trending upward year after year, and the seasonal ups and downs are strong and predictable — they are not just random noise.

3-Month Sales Forecast

I tried three different forecasting methods — a statistical model called SARIMA, a tool called Prophet, and a machine learning model called XGBoost — and compared how close each one came to actual past sales. SARIMA came out on top, with its predictions landing closest to what really happened (it was off by about 15% on average, compared to 22% for Prophet and 32% for XGBoost). I'm recommending we use SARIMA going forward. Here is what it predicts for the next three months, along with a realistic range in case actual sales land a bit higher or lower than expected:

- Month 1: I expect around $71,500 in sales, likely landing somewhere between $26,500 and $116,400.
- Month 2: I expect a dip to around $55,200, likely landing somewhere between $6,900 and $103,500.
- Month 3: I expect sales to climb back up to around $75,400, likely landing somewhere between $21,600 and $129,100.

In plain terms: I expect a dip in Month 2 before sales pick back up in Month 3. The wide ranges reflect how much our sales naturally swing month to month, so I'd treat these as a planning guide rather than an exact number.

Top 3 Unusual Sales Weeks Detected

I ran two separate checks on our weekly sales to catch weeks that didn't follow the normal pattern. Here are the three biggest and clearest ones:

- Week of March 22, 2015 — sales jumped to about $37,700, far above a normal week. This likely came from a large bulk or corporate order, possibly tied to end-of-quarter business purchasing.
- Week of December 17, 2017 — sales rose to about $25,400, in line with the pre-Christmas shopping rush.

- Week of November 4, 2018 — sales climbed to about $29,000, matching the start of the holiday shopping season.

Product Demand Segments & Stocking Strategy

I grouped our product sub-categories into three groups based on how much they sell, how fast they're growing, and how unpredictable their monthly sales are:

- High Growth, Premium Niche (Copiers): low volume but high value and growing fast. I'd keep small stock levels but reorder often, since each unit sold matters a lot to revenue.
- Low Volume, Stable Demand (Accessories, Appliances, Art, Bookcases, Envelopes, Fasteners, Furnishings, Labels, Paper, Supplies): demand here is predictable. Standard reorder points and moderate stock levels work fine, and this group doesn't need close monitoring.
- High Volume, Volatile Demand (Binders, Chairs, Machines, Phones, Storage, Tables): these sell a lot, but sales swing unpredictably and growth has flattened or dipped. I'd keep safety stock and monitor this group more closely, while avoiding overstocking since long-term growth isn't guaranteed.

Business Recommendations

- Use SARIMA forecasts as the default planning tool for monthly stock and staffing decisions, since it beat the other two models on every accuracy measure I tested (15% average error vs. 22% and 32% for the others).
- Build extra stock ahead of March, September, and November every year, since these months spike consistently across all four years of data — this isn't a one-time event, it's a repeating pattern we can plan for.
- Put tighter inventory controls and more frequent stock checks on Binders, Chairs, Machines, Phones, Storage, and Tables. These sell in high volume but swing unpredictably month to month, which is exactly where stockouts or overstock tend to happen.

Risk & Limitation to Be Aware Of

This system learns from patterns in the past, so it works best when the future looks similar to the last four years. If something unusual happens — a new competitor, a big supply chain disruption, or a sudden shift in customer demand — the forecast won't see it coming until it shows up in the data. I'd treat these numbers as a strong planning guide, not a guarantee, and revisit the forecast regularly as new sales data comes in.